Name: _____ Class: _____ No.: _____

Antibiotics（抗生素） and bacterial growth

Study sequence: explanation → worked example → guided practice → independent application → checking and correction.

All plates and data are teaching simulations. Do not operate the VL or culture MRSA（耐甲氧西林金黃葡萄球菌） physically. Shared simulated conditions: 30°C and 24 hours.

The history of antibiotic（抗生素） discovery—an unexpected observation



In 1928, Fleming noticed contamination by Penicillium（青黴菌） on plates growing staphylococci, with reduced bacterial growth nearby. He investigated an antibacterial substance produced by this fungus. Later, Florey, Chain and their team advanced the therapeutic use of penicillin（青黴素）, an antibiotic（抗生素）.

Note: Fleming’s historical work involved staphylococci, not MRSA（耐甲氧西林金黃葡萄球菌）; MRSA（耐甲氧西林金黃葡萄球菌） was first formally reported in 1961.

Image source: https://www.cbsnews.com/news/almanac-the-discovery-of-penicillin（青黴素）/

Research context

A wound infection has not clearly improved after initial antibiotic（抗生素） treatment; testing identifies MRSA（耐甲氧西林金黃葡萄球菌）. This worksheet uses static images from three independent plates to compare X, Y, Z and the matching carrier control C for visible bacterial growth.

Under the stated simulation conditions, how do X, Y and Z affect visible growth of these MRSA（耐甲氧西林金黃葡萄球菌） bacteria?

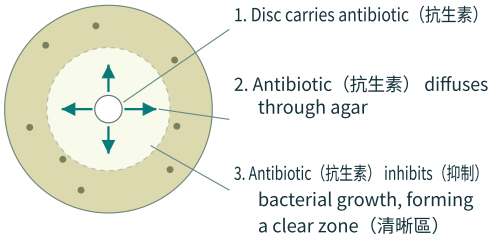
Learn the core concepts first

- Antibiotics（抗生素） mainly treat bacterial infections, not viral colds or influenza.
- Antibiotic（抗生素） diffuses outward from the disc, usually decreasing in concentration with distance. A clear zone（清晰區） forms where growth is still inhibited（抑制）. More susceptible（對抗生素敏感的） bacteria may be inhibited（抑制） at a lower concentration, allowing a larger zone.
- A clear zone（清晰區） has no visible bacterial growth; it does not prove all bacteria died. Diffusion and preparation also affect size, so the largest zone is not the best patient treatment.
- MRSA（耐甲氧西林金黃葡萄球菌） resists methicillin（甲氧西林）, not necessarily every antibiotic（抗生素）. No zone is not a clinical resistance（抗藥性） classification: this model has no clinical breakpoints and cannot be generalised to all MRSA（耐甲氧西林金黃葡萄球菌）.
- If a few resistant（具抗藥性的） bacteria already exist, antibiotic（抗生素） selection pressure（選擇壓力） favours their survival, reproduction（繁殖） and inheritance（遺傳） of resistance traits（抗藥性特徵）. Their proportion may rise and effectiveness may decline without their occupying the entire population.
- Bacteria do not purposefully produce the resistance（抗藥性） they need to adapt, and patients themselves do not develop bacterial resistance（抗藥性）. Misuse（不當使用） or overuse（過度使用） adds unnecessary selection pressure（選擇壓力）.

Complete B yourself and preserve your first answers before reading C. The examples in A are not the results for X/Y/Z/C.

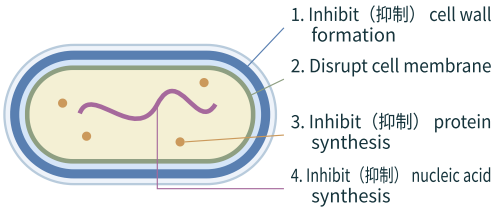
Worked examples: applying concepts to investigation and measurement

A. Experimental principle: antibiotic（抗生素） diffusion and clear-zone（清晰區） formation



Schematic, not to scale; it does not prove complete killing.

B. Structures and possible action sites



The pale outer layer represents a capsule (not present in all bacteria); blue is the cell wall, green is the cell membrane, and the purple strand is DNA. These mechanisms are not proved by this disc experiment.

Testable hypotheses and variables

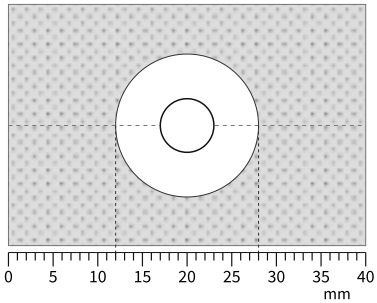
Example: compare sample P with the same carrier control for bacterium T. “With other conditions equal, P has a larger external-zone total diameter than the carrier control.” This names a comparison and a measurable outcome, but cannot prove complete killing from zone size. It is not an X/Y/Z prediction.

Example: if only disc size changes for the same sample, disc size is independent; external-zone presence and total diameter are dependent. Keep bacteria and carrier alike. For the X/Y/Z task, classify the factor actually being changed.

Control, spacing and measurement

C contains the same carrier without antibiotic（抗生素） as a comparison baseline. Call it a water control only when the carrier is water. Space discs apart and away from the edge to avoid overlap and difficult interpretation.

Example S (not X/Y/Z/C): an external zone surrounds a 6 mm disc. Its ends read 12 mm and 28 mm; total diameter = 28−12 = 16 mm, including the disc. With only a disc and no external zone, this model records 0, not 6.



Mean total diameter calculation

Example S has 12, 16 and 20 mm from three independent plates: mean = (12+16+20) ÷ 3 = 16.0 mm. Repeats check variation. First investigate spreading, contamination, disc positions and conditions, then consider additional independent tests. Never delete data for disagreeing with a hypothesis. Three is a classroom arrangement, not the only reliable number.

Complete B yourself and preserve your first answers before reading C. The examples in A are not the results for X/Y/Z/C.